Gayatri Vidya Parishad College of Engineering Autonomous

Civil Engineering | Course Code: CE201

Surveying & Geomatics - Surveying Field Methods, Theodolite & Total Station

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Overview: Bowditch balancing rule, height of instrument method, rise-and-fall levelling, trigon

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Linear Measurements & Compass Surveying
Chain and tape surveying, Magnetic declination, Whole circle bearing vs reduced bearing, Local attra

Unit 2: Levelling & Contouring
Differential levelling, Collimation and Rise-and-Fall computation methods, Reciprocal levelling, Con

Unit 3: Theodolite Traversing & Computation
Vernier transit theodolite, Measurement of horizontal and vertical angles, Gale traverse table, Bowd

Unit 4: Modern Geomatics & Curve Setting
Circular and transition curves, Rankine deflection angle method, Total Station EDM operations, GPS/G

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Reduce levels from given level book field readings using both Height of Instrument and Rise-and-Fall m
2. Calculate corrected latitudes and departures for a closed polygon traverse using Bowditch balancing me
3. Formulate the setting-out table for a simple circular curve of radius 300 m with 20 m chord increments

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* B.C. Punmia, Ashok Kumar Jain - Surveying Vol. I & II Laxmi Publications
* K.R. Arora - Surveying Vol. 1 & 2 Standard Book House

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